

Question:
An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

- Answer Choices:
- A. 3.0 amps, 4 ohms
 - B. 2.8 amps, 3.2 ohms
 - C. 3.75 amps, 3.2 ohms
 - D. 4 amps, 3 ohms
 - E. 3.33 amps, 3.6 ohms
 - F. 5 amps, 2.4 ohms
 - G. 4.5 amps, 2.67 ohms
 - H. 2.2 amps, 5.45 ohms
 - I. 2.5 amps, 4.8 ohms
 - J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>
- Use the electrical relation $P = V \cdot I$, with the battery's terminal voltage $V = 12\text{ V}$ and power $P = 40\text{ W}$. Solving gives the current:
 $I = P / V = 40\text{ W} / 12\text{ V} \approx 3.33\text{ A}$.

- By Ohm's law, the bulb's resistance is $R = V / I$. With $V = 12\text{ V}$ and $I \approx 3.33\text{ A}$:
 $R \approx 12\text{ V} / 3.33\text{ A} \approx 3.6\text{ ohms}$.

Therefore, the headlight draws about 3.33 amps and has a resistance of about 3.6 ohms. The correct choice is E.

</think>
<answer>
E. 3.33 amps, 3.6 ohms
</answer>

